

Cha Ghor

Tea estate management for the Sylhet gardens · from leaf to ledger

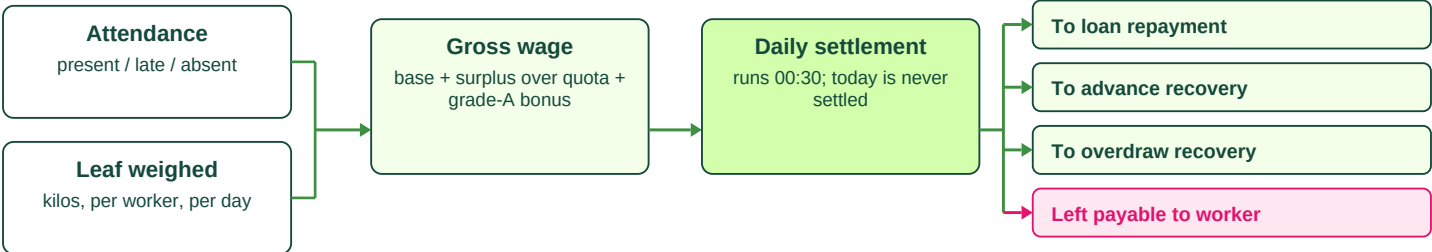
Crimson Knights

CSE 3422 / CSI 322 (C)

Software Engineering Laboratory

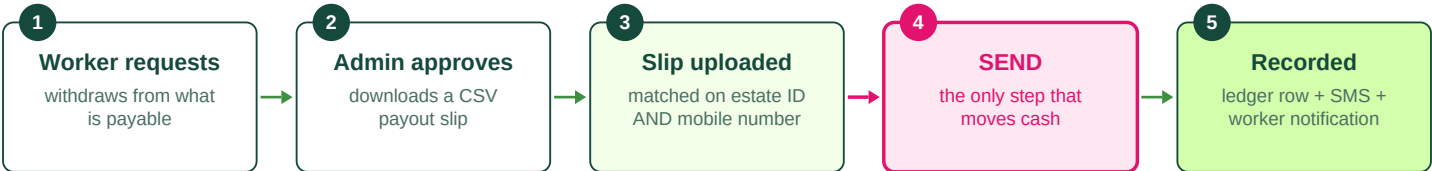
Offline-first, Bangla-language tea garden management. A field force of supervisors records attendance, leaf and wages in the garden itself — with twelve AI features that advise, but never decide a wage.

1 — THE DAILY CYCLE debt moves every day; no cash leaves the estate here



THE KEY IDEA These four always sum to what was earned. Deductions are RECORDED, not forecast — and the payslip is a statement of a settlement that already happened, not the payment itself. The worker is paid daily; the payslip only tells her what she was paid and why.

2 — THE PAYOUT approving does not pay; exactly one step moves cash



WHY THIS IS SAFE The slip chooses WHO is paid, never HOW MUCH. The amount is read from the database and the figure in the file is discarded, so an edited slip can stop a payment — it cannot cause a wrong one, redirect it to another number, or pay the same request twice. Unmatched rows are reported with a reason, never silently dropped.

3 — WHAT IS IN IT

CORE · 8 modules	AI · 12 endpoints	SECURITY
Workforce — workers, zones, fields Attendance — late pays a full day Leaf collection — weighed in the field Payroll — draft > review > approved > paid Loans & advances — recovered daily Finance — one auditable ledger Reports — cost visibility + PDF bKash payout — simulated, no live API	VISION: leaf grade suggestion, leaf health / disease ADVISORY: pluck-round advisor, weather brief, loan risk, case review LANGUAGE: Cha Bot in Bangla, SMS rewrite, loan note, report writer, worker-data extraction NO MODEL: proxy-attendance flags and rain impact are statistics, not AI	BCrypt passwords · worker PIN hashed JWT HS384, 24h; refuses to start on a weak secret 3 roles enforced per endpoint 300 req/min general, 5 login/min, answered with HTTP 429 X-Forwarded-For only from trusted proxies — closed a real bypass HSTS · CORS on API and WebSocket AI reads 5 views, read-only role

4 — ARCHITECTURE



MEASURED, NOT CLAIMED Leaf grading scored 56.7% on 97 labelled Sylhet photographs against a 51% always-guess baseline ($p = 0.15$) — no better than guessing, so the grade is suggested and never pre-filled. Leaf health has never been measured and no accuracy is claimed for it. The rate limiters are per instance, so they stop a single abusive client, not a distributed attack.